FROM THE COUNTER TO THE CLOUD

A resilient cloud migration strategy for a street-corner retail store

Case study · Methodologies, tools, standards and procedures

1. Amina's Tuesday morning

Amina opens the shutters of her corner grocery before sunrise. For twelve years the shop has run on a cash drawer, a stubborn desktop PC, a well-thumbed supplier notebook, and an Excel file she named "STOCK_FINAL_v7". Last winter a burst pipe soaked the office. Two months later a laptop vanished from behind the counter. The stock file, the supplier prices, and half her customer tabs went with it. She still served the neighbourhood that week — rice, milk, prepaid airtime — but she felt the shop wobble.

This strategy is written for her, not for a data centre. The aim is modest and human: keep the till ringing if the PC dies, protect card payments, stop losing stock counts, and give Amina a night's sleep. Cloud is the means, not the trophy.

2. Current state and the 6 Rs

The estate is tiny: a Windows PC, local Excel, paper ledgers, a standalone card terminal, WhatsApp for regulars, and a consumer-grade router. Using AWS's six common migration strategies (rehost, replatform, repurchase, refactor, retire, retain), the shop does not need to rebuild software. It needs to buy the right services and retire the fragile ones.

6 R path	Workload	Decision for the corner store
Repurchase	Till, inventory, bookkeeping	Move to cloud POS + online accounting (SaaS).
Rehost	Receipts, photos, supplier PDFs	Lift files into OneDrive / Google Drive with versioning.
Retire	Paper stock book, local Excel master	Stop after a two-week dual-run. Archive, then lock.
Retain	Cash drawer; printed price list	Keep as offline fallback when the line drops.
Replatform / Refactor	Custom apps	Not justified. No in-house code to rewrite.

This mapping follows the SME roadmap of Khan and Al-Yasiri (2015): prepare the people and contracts first, state requirements in plain language, then migrate in thin slices rather than a single "big bang" weekend.

3. Migration steps, tools and considerations

Step 1 — Assess and prepare (two weeks)

Walk the shop with Amina. List what must never stop: taking cash, taking cards, knowing what is on the shelf. Photograph the current PC, backup habits, and Wi-Fi. Score readiness: digital literacy, internet reliability, and budget. A cheap 4G failover router is a resilience control, not a luxury. Record a simple risk register: theft, flood, power cut, card fraud, staff error.

Step 2 — Choose services under a shared-responsibility model

Prefer SaaS over building servers. Shortlist a PCI DSS–certified cloud POS with offline mode (for example Square, Lightspeed, or a regional equivalent), cloud accounting (Xero or QuickBooks Online), Microsoft 365 Business Basic or Google Workspace for mail and files, and a password manager with multi-factor authentication. The shop remains responsible for staff access, device hygiene, and what data it puts in the cloud; the vendor is responsible for the platform. That split is the working procedure behind ISO/IEC 27017.

Step 3 — Pilot on one counter

Run the new POS beside the old till for ten trading days. Migrate one product category first — beverages, say — so a mistake is visible and small. Train Amina and her two assistants on three actions only: sell, refund, count stock. Skills, not tooling, decide retail cloud success (Jaka & van den Berg, 2020).

Step 4 — Cut over, then harden

Switch cards to a hosted payment page or a certified terminal so cardholder data never sits on the shop PC (PCI DSS scope reduction). Turn on daily encrypted backups and version history. Write a one-page runbook: “If the internet dies, use offline POS and the 4G dongle. If the tablet dies, use the spare. If both fail, sell cash-only from the printed list.” Review costs after 30 days.

TOOLS AND CONTROLS AT A GLANCE			
Layer	**Tool / technique**	**Resilience effect**	**Standard / procedure**
Commerce	Cloud POS + offline mode	Sales continue if the WAN fails	PCI DSS SAQ; hosted payments
Money	Xero / QuickBooks Online	Books survive a stolen PC	Daily backup; role-based access
Files & mail	M365 or Google Workspace	Versioning after flood or theft	ISO/IEC 27017 shared controls
Identity	MFA + password manager	Stops a shared-till password leak	Least privilege; joiner-leaver
Network	Fibre + 4G failover router	Second path when the street line dies	Availability target in the SLA
Recovery	One-page DR runbook + spare tablet	Staff can act without an IT firm	Test restore every quarter

4. Flowchart — how the shop actually moves

ASSESS	SELECT	PILOT	CUT OVER	HARDEN	REVIEW
Walk the shop. List must-not-fail work. Score internet, skills, cash.	Repurchase SaaS. Check PCI, SLA, offline mode, local support.	One category, ten days, dual till. Train three tasks only.	Retire Excel master. Point cards at hosted / certified path.	MFA, backups, 4G failover, least privilege, runbook.	30-day cost check. Quarterly restore test. Adjust licences.

→ If the pilot fails, stay dual-run and change vendor. Do not force a cutover. ←

5. Standards and procedures that actually bind the shop

NIST SP 800-145 still gives the cleanest definition of what Amina is buying: on-demand, pooled, rapidly provisioned services. For implementation she does not need a full ISO 27001 certificate on day one. She does need procedures that those standards describe in smaller language:

- PCI DSS (and the PCI SSC cloud guidance): never store full card data on the shop PC; prefer a certified terminal or hosted checkout; complete the correct Self-Assessment Questionnaire for merchant volume.

- ISO/IEC 27017: write down who does what — vendor patches the POS platform; Amina revokes a leaver's login the same afternoon.
- NIST Cybersecurity Framework functions, used as a weekly habit: Identify assets, Protect with MFA and backups, Detect odd refunds, Respond with the runbook, Recover by restoring yesterday's stock file.
- Contract procedure: read the SLA for uptime and data-export rights before paying the first month. Cloud lock-in is a business risk, not only a technical one.

Retail SME research keeps returning to the same enablers: relative advantage, trialability, top-management (here: owner) support, and trust in vendors (Udakumbura, Wijayanayake & Jayasinghe, 2025). The pilot step is there because theory says Amina will only believe the cloud when she has sold a crate of water on it.

6. What "resilient" means on a street corner

Resilience is not three availability zones in a lecture slide. It is Amina selling bread after the municipal cable is cut. Design for that: SaaS vendors who already replicate across regions; a POS that queues tickets offline; a second network path; a spare tablet in a dry box above the flood line; a restore test every quarter; and people who have practised the runbook once, on a quiet Monday. Cost stays proportional — a few tens of dollars a month — so the strategy can survive a slow season.

If the pipe bursts again, the stock file is already somewhere the water cannot reach.

References

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